IoT Enabled Pest Detection and Disease Management System for Agriculture 4.0

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Abstract. Pest and diseases have been major threats in agricultural productivity since early ages. It causes significant crop damage. One of the critical challenges in agriculture is the effective detection of pests and disease. Traditionally huge amounts of harmful chemicals are used. This research can bring revolutionary impact on pest detection and disease management. The system collects real time data from fields utilizing an ESP32 microcontroller which includes Wi-Fi functionality which will enable seamless sensor data processing and connectivity and where Wi-Fi is unavailable SIM900 GSM module is used to connect through cellular data. A BME280(temperature, humidity), BH1750(light-intensity), MQ-135(air-quality) and Capacitive Soil Moisture Sensor. GPS (Neo-6M) is used for precise location. 18650 Li-ion batteries are the primary source of power and to recharge solar panels are used. Arduino cloud receive and analyze the collected data and compare with standard value range to identify patterns, predict pest infestation and detect disease at an early stage. IoT platform dashboard and SMS will send notification to the farmer about any abnormalities. 90% accuracy is achieved in field testing. This system achieved the key goals of Agriculture 4.0 by promoting sustainable eco-friendly practice, reducing use of chemicals and preserving biodiversity.

Keywords: Disease Management · IoT in Agriculture · Pest Detection · Real-time Monitoring · Smart Farming · Sustainable Agriculture.

1 Introduction

A lot of people throughout the world depend on farming for food, raw materials and jobs. It is the backbone of many economies. But pests and plant diseases are becoming a more and bigger danger to current farming methods. These problems cause crops to yield less and be of lower quality. In the past, farmers used to look for pests by hand or by looking for signs of them. They often used a lot of chemical pesticides as a way to keep bugs away. This method may help in the

short run, but it harms the environment, raises production costs, and exposes customers to dangerous chemical residues.

In places like Asia, where more than 90% of the world's rice is grown, pests and insects cause an average of 20% production loss each year [1][2]. Studies have also found that more than 29% of vegetable samples examined in the last few years had pesticide residues on them, which is bad for people's health and raises questions about food safety [3]. Also, the wrong application of pesticides causes bigger public health problems. About 250,000 to 370,000 people die each year from eating pesticides on purpose [4].

Agriculture 4.0 is a new way of farming that uses smart technologies [5]. It could help solve these problems. It stresses precision agriculture, data-driven decisions, and ecologically sensitive methods [6]. The Internet of Things (IoT) is one of the most important technologies in this trend [7]. It lets sensors and gadgets talk to each other so that environmental conditions may be monitored all the time [8]. With real-time data at their fingertips, farmers can move from reactive responses to proactive pest and disease management . This study suggests an IoT-enabled pest detection and disease management system that can keep an eye on important environmental factors, spot any risks early and cut down on the overuse of pesticides in response to these needs . The system includes smart sensors, GPS for location tagging, GSM-based connectivity and cloud-based data analytics . It offers both automatic notifications and user-driven queries via SMS, ensuring accessible even in rural places without internet connectivity [8]. This research intends to evaluate the system's performance across diverse agricultural situations by measuring its detection accuracy, response time and reliability. Ultimately, the system contributes to the goals of sustainable agriculture by supporting smart farming methods that promote crop health, limit chemical usage, and support long-term environmental balance.

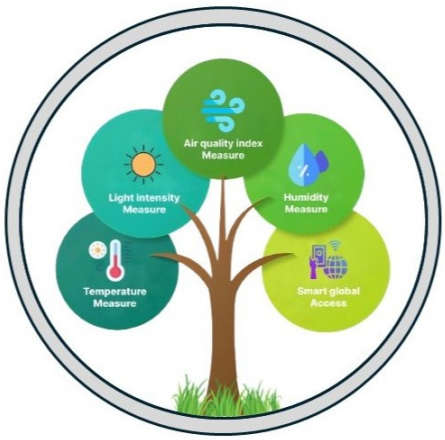

Fig. 1. Feature of IoT Enabled Pest Detection and Disease Management System.

2 Related Work

The integration of Internet of Things (IoT) technologies into agriculture has revolutionized pest and disease monitoring by enhancing the efficiency, accuracy and sustainability of farming practices. Researchers have explored numerous strategies combining sensors, communication modules and data-driven techniques.

Chougule et al. (2016) emphasized that IoT is vital for integrated pest management, involving biological, chemical and cultural tools to improve yield outcomes [6]. Materne et al. (2018) implemented wireless sensor networks (WSNs) to track eight key environmental parameters correlated with pest outbreaks and applied algorithms like KNN and Random Forest for prediction [7]. Gao et al. (2020) used UAVs with spectral cameras for image-based analysis of pest activity and proposed a wind-assisted flight mode to extend operational time [8]. Doe et al. (2023) introduced a Parallel and Distributed Simulation Framework (PDSF) that reduces GPU load and ensures real-time data delivery, even during failures [9]. Azfar et al. (2023) presented a drone-based spraying system integrated with motion detection sensors and decision-making algorithms to control cotton pests [10].Ahmed et al. (2024) applied CNN models to detect oriental fruit flies using smart image traps [11]. Ali et al. (2024) used a combination of sound and thermal sensing with DenseNet to identify pests and trigger ultrasonic repellents [12]. Wang et al. (2024) developed a coordinated IoT-based hardware-software system for both greenhouses and open fields [13]. Ibrahim et al. (2025) used a ResNet50 model to classify plant leaf diseases with 99.8% accuracy based on an augmented image dataset [14].

In contrast, our research focuses on affordability, simplicity and real-world usability in rural and off-grid contexts. Our solar-powered, low-cost prototype provides real-time sensor data via both cloud and SMS, ensuring accessibility in areas without internet. It uniquely supports user-triggered SMS responses, plug-and-play installation and automatic switching between GSM and Wi-Fi. Furthermore, one device can represent 400–500 m^2 in uniformly conditioned fields, offering cost-effective wide-area coverage. These features make our system novel compared to existing solutions that often rely on complex models, expensive drones or require high-speed internet. Our design prioritizes scalability, energy autonomy, and ease of deployment, making it well-suited for sustainable and precision agriculture under the vision of Agriculture 4.0.

3 Methodology of The System

3.1 Architecture of the System

The IoT-enabled pest detection and disease management system is developed to provide real-time monitoring and smart decision-making support for contemporary agriculture, following the goals of Agriculture 4.0. The system employs smart sensors, a low-power edge device and GSM-based communication to assure flawless functioning even in rural farming areas with poor internet connection (see Fig. 2).

At the center of the system is an edge gadget, powered by a rechargeable battery, which is charged using a solar panel. This design makes the system energy-efficient and suited for off-grid agricultural lands. The edge gadget collects data from many sensors located around the farm. These include a temperature and humidity sensor to track environmental conditions, an air quality sensor to detect harmful gases that may signal pests or disease, a soil moisture sensor to monitor water levels in the soil, a light sensor to measure sunlight exposure and a GPS module to tag the data with its location. This modular architecture allows the system to adapt to diverse crops and field conditions. After gathering the data, the edge device checks if a Wi-Fi connection is accessible. If Wi-Fi is present, the system uses MQTT over TLS to securely communicate the data to the Arduino IoT Cloud, guaranteeing that all communication is encrypted and protected. The cloud stores and processes the data, which may subsequently be accessed through a connected mobile application to view trends and receive alerts. If Wi-Fi is not available, the system resorts to using a GSM module with a SIM card. Through UART connectivity and AT commands, the edge device provides data to the GSM module. This allows for two-way SMS communication users can send an SMS request to the system and it replies with current sensor values. Also, if sensor readings reach preset levels that may signal pest outbreaks or disease danger, the system sends automatic SMS alerts to the user's phone.

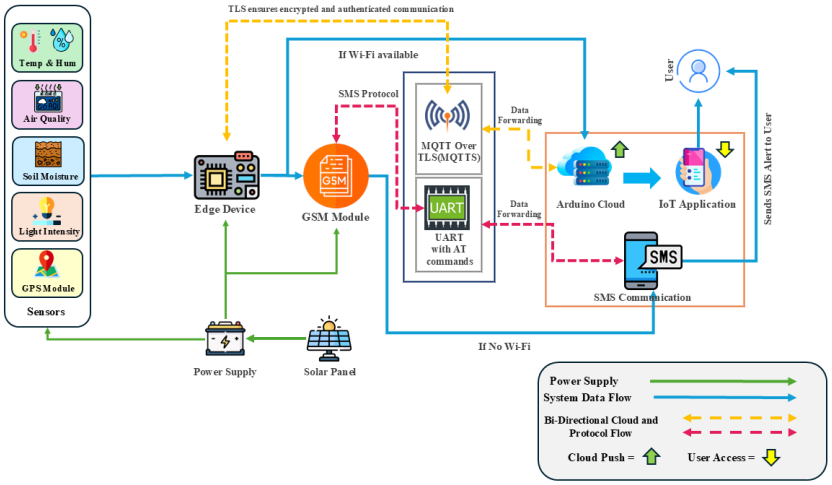

Fig. 2. Block Diagram of IoT-Based Pest and Disease Monitoring System.

The IoT smartphone application allows customers real-time access to sensor data and messages, allowing farmers to respond swiftly and reduce crop damage. This reliable and self-powered arrangement allows sustainable and smart farming, even in locations without internet or energy.

3.2 Workflow of the System

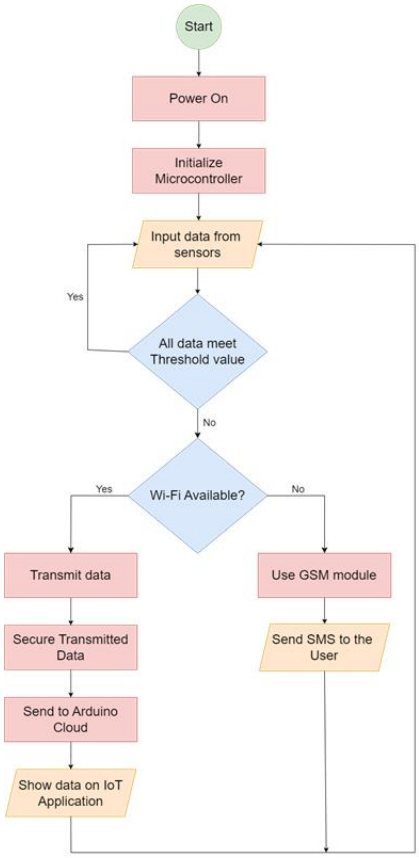

Fig. 3. Workflow of the Pest Detection and Disease Management System.

The workflow of the system (see Fig. 3) begins with the system powering on and then initializes the ESP32 microcontroller with the connected sensors. After activation, the system moves into the data collection phase. It receives real time environmental and soil data inputs from the ground from the sensors attached to the edge device. At data processing phase, the ESP32 processes data and compares them with the predefined thresholds for each parameter such as temperature, humidity, gas levels, light intensity and soil moisture.

It will continuously receive data until it exceeds the satisfactory threshold value. When it receives any abnormal value the system will proceed to check for Wi-Fi availability. If Wi-Fi is available, it encrypts the collected data. Then the encrypted data is sent to the Arduino cloud for further calculation. After uploading to the cloud, users can monitor real-time environmental conditions and

sensor readings. When abnormality is detected, the user receives notification. In case of high humidity levels, increased levels of gasses or lower soil moisture, the system goes into an alert mode. When Wi-Fi is not available, the system uses the GSM SIM900 module to directly send SMS alert to the user. The SMS notify about the location, temperature, humidity, light intensity, air quality and soil moisture. These alerts are practical information upon which quick response can be taken in preventing pest infestation or disease breakouts, ensuring effective crop management.

3.3 Stimulation of the System Circuit

To develop and verify the prototype system, both simulation tools and real-world breadboard testing were used. This approach was necessary because some components, especially sensors, did not produce accurate results in simulation software alone. The final simulation setup is illustrated (see Fig. 4). This represents an integrated IoT-based system using the ESP32 microcontroller as the central control unit.

The system is powered by two 18650 lithium-ion batteries, which together deliver up to 8.2V when completely charged. A solar panel is added to charge the batteries so that they can be used in remote farming locations. A TP4056 charging module with built-in protection is used to safely manage both charging and discharging. To avoid reverse current from flowing back into the solar panel during the night, a 1N5819 Schottky diode is added. Since the ESP32 requires 5V input, a buck converter is used to step down the battery voltage and connect it to the VIN pin of the ESP32.The MQ-135 gas sensor can detect harmful gases such as ammonia (NH_3), which may indicate the presence of pests or decomposing organic waste. Since the ESP32 does not have a 5V output pin, a buck converter is used to directly power the sensor's V_{CC}. The sensor's analog output (AO) is connected to GPIO34 (D34), an analog-to-digital input pin on the ESP32, which measures voltage levels corresponding to gas concentration.To monitor temperature and pressure, the BME280 sensor is linked using four pins: VCC (attached to the ESP32's 3.3V output), GND (to the common ground) and the I2C communication lines SDA and SCL, connected to GPIO21 (D21) and GPIO22 (D22) correspondingly. The BH1750 light intensity sensor is also attached to these same I2C lines without issue, as both sensors utilize distinct I2C addresses. For example, the BME280 commonly utilizes address 0x76 or 0x77, whereas the BH1750 uses 0x23 or 0x5C. This allows the ESP32 to talk with each sensor individually over the same bus. The BH1750 helps detect sunlight levels in lux, which is crucial for ensuring crops receive adequate light and for observing environmental variables that can influence insect activity. A capacitive soil moisture sensor is included to assess the moisture level in the soil, helping predict plant stress and pest risks. This sensor works by detecting changes in the soil's dielectric property. It can run on 3.3V or 5V and its analog output is connected to GPIO35 (D35) of the ESP32. To track the device in large fields, the NEO-6M GPS module is used. It operates on 5V from the buck converter and communicates via UART using TX and RX pins connected to the ESP32. Finally,

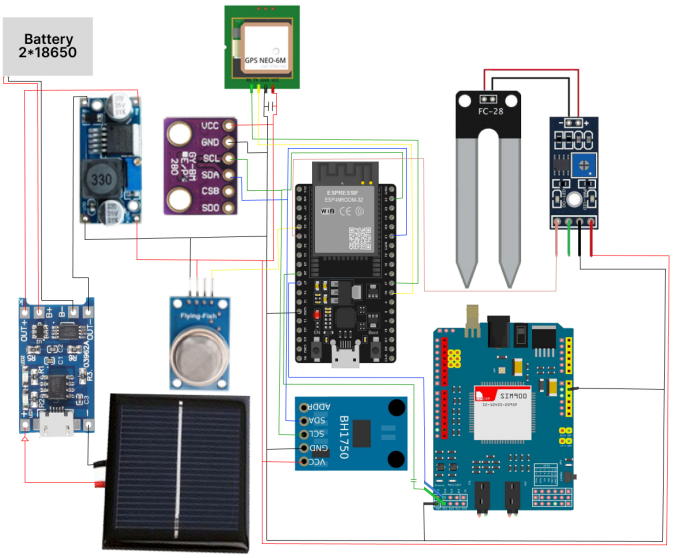

Fig. 4. Simulated Circuit of the IoT-Based Pest Monitoring System.

a SIM900 GSM module is integrated where there is no access to the internet where SMS messages are sent. It also operates at 5V using buck converter and TX and RX are connected to pin GPIO26 and GPIO27 of the ESP32. It is able to pull more than 2A current during transmission so a capacitor is used to smooth the power supply, a 1000uF 25V capacitor is placed between VCC and GND. The simulation indicates an entire functional prototype and its ability to string all components together to allow real-time monitoring of the environment, detection of the pests and placement as the basis of practical implementation in smart agriculture systems.

3.4 Experimental Setup

To confirm the performance of the proposed system, a functioning prototype was created and tested in multiple phases (see Fig. 5). The development procedure began with constructing the components on a breadboard to evaluate circuit integrity and operation (see Fig. 5(a)).This stage allowed flexible debugging and alignment with the previously simulated circuit design.

Upon successful breadboard testing, the components were converted to a permanent arrangement using a Veroboard, ensuring solid electrical connections. To protect the hardware in external conditions, the system was covered in a sturdy PVC shell (see Fig. 5(b)).The enclosure was rendered weather-resistant by the application of waterproof coatings and gaps were sealed using industrial-grade adhesive to prevent water intrusion during rains and muddy conditions.

The system was powered by 18650 Li-ion batteries, which were rechargeable via integrated solar panels to provide continuous off-grid operation. The ESP32

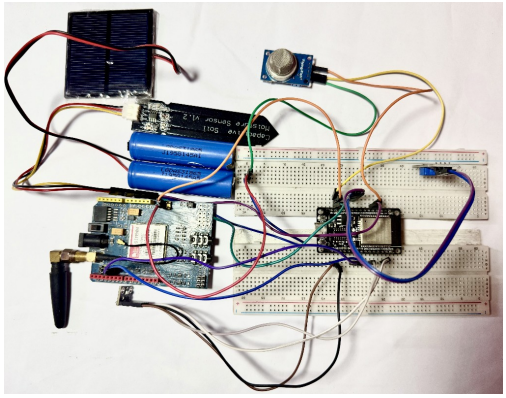

Fig. 5(a). Prototype of the Smart Pest and Crop Health Monitoring System

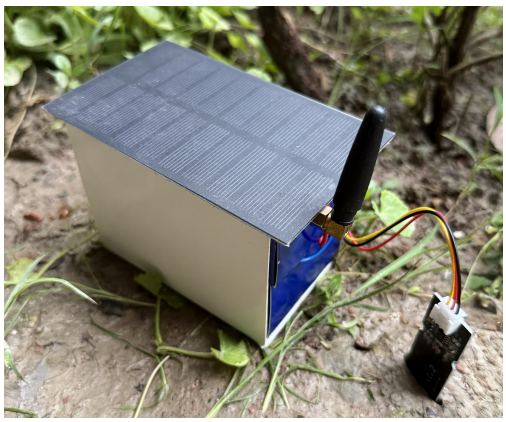

Fig. 5(b). In field demonstration of prototype

Fig. 5. Development and Deployment Phases of the IoT-Based Prototype

microcontroller was programmed using the Arduino IDE and numerous libraries were utilized to provide seamless operation of all components. The ArduinoIoT-Cloud library provided real-time synchronization of sensor and GPS data with the Arduino IoT Cloud dashboard. The Arduino_ConnectionHandler library coordinated automatic Wi-Fi reconnections for uninterrupted cloud connectivity. Environmental data were acquired using the Adafruit_BME280 (for temperature and humidity) and BH1750 (for light intensity) sensors with support from the Adafruit_Sensor and Wire libraries for I2C connection. For geolocation, the TinyGPS++ library decoded latitude and longitude values from the Neo-6M GPS module. To avoid conflict with USB connection, the HardwareSerial library was utilized to configure a separate UART interface (Serial1) for specialized GPS data gathering.

Following software integration, the prototype was tested under indoor simulated environmental conditions to determine accuracy and durability. Upon confirmation of stable performance, the completed waterproofed devices were placed in actual agricultural areas (see Fig. 5(b)). A total of three identical prototypes were created and put in different field zones to maximize data collecting coverage and enable thorough field-level validation.

4 Result and Analysis

To test the real-world effectiveness of the prototype IoT-based pest and disease monitoring system, a field study was undertaken over a three-month period across three agricultural locations in Bangladesh. The selected plots were farmed with eggplant, tomato, and ladies' finger crops that are commercially significant and extremely sensitive to pest and disease outbreaks in tropical climes. Each location was supplied with a deployed prototype setup to monitor temperature, humidity, air quality (by MQ-135), light intensity, soil moisture, and GPS coordinates. Data was continually uploaded to the Arduino IoT Cloud for real-time display and analysis (see Fig. 6(a)). In locations without Wi-Fi connectivity, GSM-based SMS notifications delivered real-time updates to farmers (see Fig. 6(b)). Additionally, when farmers require on-demand information, the system supports two-way communication, allowing users to request current sensor data via SMS and receive an instant response (see Fig. 6(c)). The cloud dashboard permitted real-time threshold comparisons and automatically issued notifications when values exceeded prescribed agronomic limits. Furthermore, the system's energy consumption was tracked to evaluate long-term viability. The reliability of sensor data was verified through periodic calibration checks. Farmers provided positive feedback on the improved speed of pest identification. Threshold levels were fine-tuned based on field expert input during the study. Alerts helped prevent severe outbreaks, reducing pesticide use. Finally, the system demonstrated potential scalability for larger farming operations.

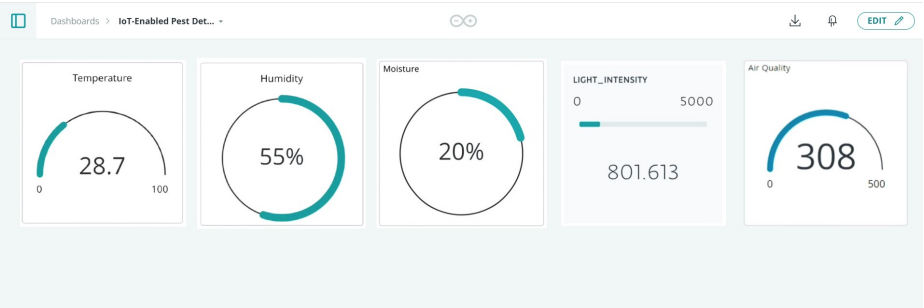

Fig. 6(a). Sensor Data Visualization Interface on Arduino Cloud.

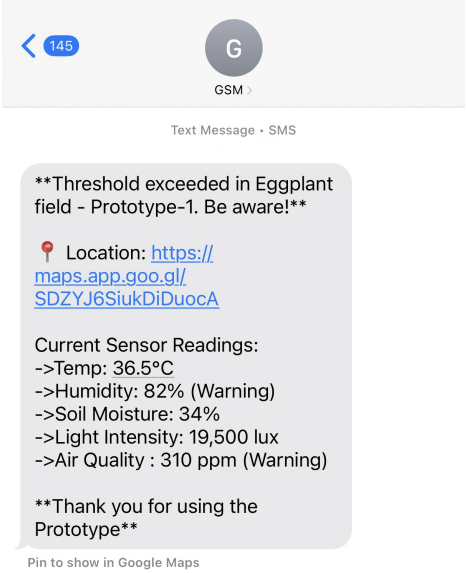

Fig. 6(b). SMS Alert Triggered via GSM Module Upon Threshold Exceedance.

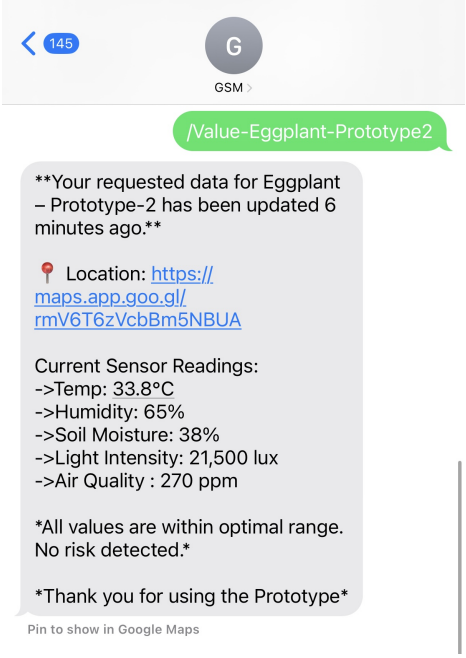

Fig. 6(c). Two-Way SMS Communication Between User and GSM-Enabled Prototype.

Table 1: Optimal Environmental Conditions for Selected Vegetables

Parameter	**Eggplant**	**Tomato**	**Ladies' Finger**
Ideal Temperature (°C)	20–37°C	18–32°C	22–40°C
Ideal Humidity	50–70%	50–70%	40–60%
Soil Moisture	30–45%	25–40%	20–35%
Light Intensity	10,000–25,000 lux	15,000–25,000 lux	15,000–30,000 lux
Air Quality (MQ-135)	$<$ 300ppm	$<$ 300ppm	$<$ 300ppm

Table 2: Eggplant Field Data and Pest Risk Analysis

Parameter	**Observed Value**	**Exceeded?**	**Possible Pests (Bangladesh)**
Ideal Temperature (°C)	42°C	Yes	Fruit & Shoot Borer (Leucinodes orbonalis)
Ideal Humidity	65%	No	–
Soil Moisture	28%	Yes	Fungal pathogens, root rot
Light Intensity (lux)	19,000	No	–
Air Quality (MQ-135)	210	No	–

Table 3: Tomato Field Data and Pest Risk Analysis

Parameter	**Observed Value**	**Exceeded?**	**Possible Pests (Bangladesh)**
Ideal Temperature (°C)	34°C	Yes	Whiteflies, Leaf miners
Ideal Humidity	76%	Yes	Powdery mildew, Early blight
Soil Moisture	27%	No	–
Light Intensity	21,000	No	–
Air Quality (MQ-135)	270	No	–

Table -1 outlines the best environmental conditions for the selected crops, determined from agronomic literature and expert consultation. These criteria were used to examine if environmental changes created a danger of insect outbreaks. During the test period, sensor data repeatedly exceeded permissible limits particularly in temperature, humidity and air quality. These threshold breaches were typically found during peak sunshine hours or in settings with inadequate air circulation.

Tables -2, 3, and 4 indicate individual occurrences where environmental circumstances reached threshold levels and were followed by observed pest activity. In these situations, either pest signs or actual pests were discovered within 3–5 days of alarm production. This substantial link demonstrates the system's utility in early warning and quick response. The findings demonstrate that sustained exposure to severe climatic conditions such as excessive temperature, high humidity, poor air quality and inadequate soil moisture significantly increases pest sensitivity across all three crops.

Table 4: Lady's Finger Field Data and Pest Risk Analysis

Parameter	**Observed Value**	**Exceeded?**	**Possible Pests (Bangladesh)**
Ideal Temperature (°C)	38°C	No	–
Ideal Humidity	78%	Yes	Aphids, Jassids, Powdery mildew
Soil Moisture	19%	Yes	Weak plant immunity
Light Intensity	17,500	No	–
Air Quality (MQ-135)	440	Yes	Whiteflies, Leafhoppers

In the eggplant field, temperatures above 37°C and humidity levels above 80% generated excellent circumstances for the development of fruit and shoot borers and whiteflies. These pests were frequently seen within 3–5 days following threshold breaches. Preventive methods may include enhancing field ventilation, timely watering to alleviate temperature stress and placing pheromone traps. In the tomato plot, comparable threshold breaches favored infestations by leaf miners, whiteflies, and fungal diseases such as early blight. Pest symptoms occurred within 2–4 days of continuous unfavorable circumstances.Effective management strategies include mulching, neem-based biopesticides and maintaining proper spacing for enhanced ventilation.

In the ladies' finger field, excessive humidity mixed with low soil moisture impaired plant immunity, favoring outbreaks of aphids, jassids and powdery mildew. These pests were spotted within 3–6 days after data alerts. Control techniques may include adjusting irrigation schedules, maintaining regular soil moisture, and adding organic repellents. Across all sites, maintaining optimal conditions particularly soil moisture between 30–40%, humidity below 70% and air quality below 300 ppm contributed to a considerable reduction in pest occurrence. The system's early detection capacity permitted prompt interventions, decreasing both pesticide consumption and crop losses.

In all, nearly 30 automatic notifications were created. Of these, 26 were validated by field observation, giving in an estimated accuracy percentage of 86.7%. Farmers remarked that fast notifications via SMS and cloud dashboards enabled preventive steps that avoided damage.

These data demonstrate the efficacy of the suggested approach under real-world settings. The system offers a cost-effective, eco-friendly and IoT-integrated method to pest and disease monitoring aligning with the aims of Agriculture 4.0 by fostering data-driven, sustainable agricultural practices.

5 Conclusion

This research presents a low-cost, solar-powered and GSM-integrated IoT-based pest and disease monitoring system designed to support sustainable agriculture in rural and connectivity-limited regions. By continuously collecting environmental data from key parameters such as temperature, humidity, air quality,

soil moisture and light intensity, the system provides real-time field insights and early warnings through cloud dashboards and SMS alerts. Field testing across three different agricultural plots demonstrated its potential to reduce pesticide misuse, support organic farming, and align with the broader goals of Agriculture 4.0.

However, despite its effectiveness, several limitations were identified during implementation. The current prototype does not perform real-time data analysis, it only monitors and transmits raw sensor data. This makes it less useful for farmers who lack technical knowledge or the ability to interpret environmental readings. To fully empower users, future versions must integrate AI or machine learning models for automatic detection, prediction and recommendation features. Another shortcoming is the sensor accuracy in certain cases, sensor readings were slightly inconsistent, making it difficult to evaluate precise environmental conditions. Moreover, the system currently relies on third-party cloud platforms like Arduino IoT Cloud, which limits scalability and data ownership. Developing a dedicated IoT platform would allow for better customization, security, and data control. The power supply also poses a constraint. Although the battery backup allows operation for up to 28 hours, continuous cloudy or rainy weather for more than two days can interrupt functionality. This affects reliability in off-grid scenarios. Enhancing energy storage capacity or using hybrid charging systems may be necessary for uninterrupted performance.

In future work, the prototype will be improved with advanced sensors, energy-efficient hardware and AI-powered analytics to offer automated pest detection and decision support. The ultimate goal is to build a reliable, intelligent, and accessible tool that helps farmers reduce chemical usage, protect crops and deliver safe, organic food to consumers. This innovation represents a step forward in reducing the burden on farmers and promoting healthier, environmentally responsible agriculture.

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Disclosure of Interests The authors have no competing interests to declare that are relevant to the content of this article.

References

1. Bangladesh Rice Journal: Insect pests cause 20% average yield loss in Asia where more than 90% of the world's rice is produced. Bangladesh Rice Journal (2021)
2. Bangladesh Rice Journal: A Way Forward to Combat Insect Pest in Rice. Bangladesh Rice Journal 25(1) (2021). https://doi.org/10.3329/brj.v25i1.55176
3. El-Sheikh, E.A., et al.: Health risk assessment of pesticide residues in vegetables collected from Dakahlia, Egypt. J. Plant Prot. Res. 61(1) (2021). https://doi.org/10.24425/jppr.2021.137951

4. Gunnell, D., et al.: The global distribution of fatal pesticide self-poisoning: Systematic review. Lancet Global Health (2007). https://pmc.ncbi.nlm.nih.gov/articles/PMC8394312/
5. Klerkx, L., Jakku, E., Labarthe, P.: A review of social science on digital agriculture, smart farming and agriculture 4.0: New contributions and a future research agenda. NJAS - Wageningen Journal of Life Sciences 90-91, 100315 (2019)
6. FAO: Digital agriculture: Farmers in the driver's seat. (2022). https://www.fao.org/digital-agriculture/en/, last accessed 2025/06/30
7. Wolfert, S., Ge, L., Verdouw, C., Bogaardt, M.-J.: Big Data in Smart Farming – A review. Agric. Syst. 153, 69–80 (2017)
8. Jayashree, V., Vaidehi, V., Kumaravel, A.: Internet of Things (IoT) for precision agriculture application. In: 3rd Int. Conf. on Communication and Electronics Systems (ICCES), pp. 1076–1080 (2018)
9. Chougule, A., Jha, V.K., Mukhopadhyay, D.: Using IoT for integrated pest management. In: Int. Conf. on Internet of Things and Applications (IOTA), pp. 17–2. Pune, India (2016)
10. Materne, N., Inoue, M.: IoT Monitoring System for Early Detection of Agricultural Pests and Diseases. In: 12th SEATUC, Yogyakarta, Indonesia, pp. 1 (2018)
11. Gao, D., Sun, Q., Hu, B., Zhang, S.: A Framework for Agricultural Pest and Disease Monitoring Based on IoT and UAVs. Sensors 20(5), 1487 (2020)
12. Doe, J., Smith, A., Lee, B.: Control of pests and diseases in plants using IoT technology. J. Smart Agric. (2023)
13. Azfar, S., Nadeem, A., Ahsan, K., et al.: IoT-Based Cotton Plant Pest Detection and Smart-Response System. Appl. Sci. 13(3), 1851 (2023)
14. Ahmed, S., Marwat, S.N.K., Brahim, G.B., et al.: IoT based intelligent pest management system for precision agriculture. Sci. Rep. 14, 31917 (2024)
15. K. I. H. Turjo, M. Ahmed, K. Redwan, M. F. A. Al Sohan, A. Nahar and M. S. Mahmood, "IoT-Enabled Smart Air Traffic Control for Drones in Urban Areas: AI-Driven Route Optimization and Energy Management," 2025 2nd International Conference on Advanced Innovations in Smart Cities (ICAISC), Jeddah, Saudi Arabia, 2025, pp. 1-6, doi: 10.1109/ICAISC64594.2025.10959669.
16. Ali, M.A., Dhanaraj, R.K., Kadry, S.: AI-enabled IoT-based pest prevention and controlling systems using sound analytics in large agricultural fields. Comput. Electron. Agric. 220, 108844 (2024)
17. S. S. Akter, M. M. Hossain Munna, K. Redwan, M. Ahmed and M. F. A. Al Sohan, "IntelliGuard: An AI-Powered Threat Detection System for Smart Home Operations," 2025 2nd International Conference on Advanced Innovations in Smart Cities (ICAISC), Jeddah, Saudi Arabia, 2025, pp. 1-6, doi: 10.1109/ICAISC64594.2025.10959362.
18. Wang, Z., Qiao, X., Wang, Y., Yu, H., Mu, C.: IoT-based system of prevention and control for crop diseases and insect pests. Front. Plant Sci. 15, Article 1323074 (2024)
19. A. Nahar, M. F. A. Al Sohan, K. Redwan and M. Ahmed, "Advanced IoT-Enabled Forest Management System for Wildfire Detection and Ecosystem Protection: A Case Study in the Sundarbans," 2025 2nd International Conference on Advanced Innovations in Smart Cities (ICAISC), Jeddah, Saudi Arabia, 2025, pp. 1-6, doi: 10.1109/ICAISC64594.2025.10959377.
20. Ibrahim, A.S., Mohsen, S., Selim, I.M., et al.: AI-IoT based smart agriculture pivot for plant diseases detection and treatment. Sci. Rep. 15, 16576 (2025)

21. M. A. Sarker Emon, M. F. Abdullah Al Sohan, M. M. Hossain Munna, M. Ahmed and K. Redwan, "CityShield: An IoT-Driven and AI-Based Threat Detection System for Smart City Operations," 2025 2nd International Conference on Advanced Innovations in Smart Cities (ICAISC), Jeddah, Saudi Arabia, 2025, pp. 1-6, doi: 10.1109/ICAISC64594.2025.10959442.
22. M. Ahmed, K. Redwan, M. F. A. Al Sohan and A. Nahar, "Next-Gen IoT-Driven Framework for Real-Time Marine Ecosystem Support and Management," 2024 10th International Conference on Computing, Engineering and Design (ICCED), Jeddah, Saudi Arabia, 2024, pp. 1-6, doi: 10.1109/ICCED64257.2024.10982754.
23. K. Redwan, M. Ahmed, K. I. H. Turjo, M. M. Hossain Munna, S. S. Akter and M. F. A. Al Sohan, "Smart IoT-Driven Shopping Assistance System for Cognitive Impairment Patients: Ensuring Safe and Independent Shopping Experiences," 2024 10th International Conference on Computing, Engineering and Design (ICCED), Jeddah, Saudi Arabia, 2024, pp. 1-6, doi: 10.1109/ICCED64257.2024.10983645.